

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATA SCIENCE AND MINING LAB

PCCSL508 MACHINE LEARNING LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Linear Regression with One Variable — California Housing Dataset	CO1,CO3,CO4
2	Polynomial Regression — Auto MPG Dataset	CO1,CO3,CO4
3	Ridge and Lasso Regression — Diabetes Dataset	CO1,CO3,CO4
4	Logistic Regression — MLE and MAP Estimation on Breast Cancer Wisconsin Dataset	CO1,CO2,CO3,CO4
5	MLE and MAP for Multinomial Distribution — 20 Newsgroups Dataset	CO1,CO2,CO3,CO4
6	Logistic Regression with and without Feature Scaling — Pima Indians Diabetes Dataset	CO1,CO3,CO4
7	Naive Bayes Text Classification — 20 Newsgroups Dataset (Multinomial vs Bernoulli)	CO1,CO2,CO3,CO4
8	K-Nearest Neighbours (KNN) for Image Classification — Fashion MNIST Dataset	CO1,CO3,CO4
9	Decision Tree (ID3) for Customer Segmentation — Online Retail Dataset	CO1,CO3,CO4
10	Logistic Regression vs Decision Trees — Adult Income Dataset	CO1,CO3,CO4
11	Linear Support Vector Machine (SVM) — Iris Dataset	CO1,CO3,CO4
12	SVM Kernel Comparison (Linear, Polynomial, RBF) — Fashion MNIST Dataset	CO1,CO3,CO4

13	Multilayer Feed-Forward Network (MLP) — Wine Quality Dataset	CO1,CO2,CO3,CO4
14	Activation Function Comparison (Sigmoid, ReLU, Tanh) — MNIST Dataset	CO1,CO2,CO3,CO4
15	Hyperparameter Tuning of a Neural Network — Fashion MNIST Dataset	CO1,CO2,CO3,CO4
16	Hierarchical vs K-Means Clustering — Mall Customers Dataset	CO1,CO3,CO4
17	K-Means Clustering — Digits Dataset	CO1,CO3,CO4
18	Bootstrapping and Cross-Validation — Iris Dataset	CO1,CO2,CO3,CO4
19	Bagging and Boosting Ensemble Methods — Titanic Dataset	CO1,CO3,CO4
20	Bias-Variance Tradeoff via Polynomial Regression — Boston Housing Dataset	CO1,CO3,CO4

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATA SCIENCE AND MINING LAB

CSL411 COMPILER LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Design and implement a lexical analyzer using C language to recognize all valid tokens in the input program. The lexical analyzer should ignore redundant spaces, tabs, newlines and comments.	CO1
2	Implement a Lexical Analyzer for a given program using LEX Tool.	CO1
3	Write a LEX program to display the number of lines, words and characters in an input text.	CO1
4	Write a LEX program to convert the substring abc to ABC from the given input string.	CO1
5	Write a LEX program to find the total number of vowels and consonants from the given input string.	CO1
6	Generate a YACC specification to recognize a valid arithmetic expression that uses operators +, -, *, / and parenthesis.	CO2
7	Generate a YACC specification to recognize a valid identifier which starts with a letter followed by any number of letters or digits.	CO2

8	Implementation of Calculator using LEX and YACC.	CO2
9	Convert the BNF rules into YACC form and write code to generate an Abstract Syntax Tree (AST).	CO2
10	Write a program to find ϵ -closure of all states of any given NFA with ϵ transition.	CO3
11	Write a program to convert NFA with ϵ transition to NFA without ϵ transition.	CO3
12	Write a program to convert NFA to DFA.	CO3
13	Write a program to minimize any given DFA.	CO3
14	Write a program to find FIRST and FOLLOW of any given grammar.	CO4
15	Design and implement a Recursive Descent Parser for a given grammar.	CO4
16	Construct a Shift Reduce Parser for a given language.	CO5
17	Write a program to perform constant propagation.	CO5
18	Implement Intermediate Code Generation for simple expressions.	CO5
19	Implement the back end of the compiler which takes the three-address code and produces 8086 assembly language instructions (move, add, sub, jump, etc.).	CO5

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATA SCIENCE AND MINING LAB

PCCSL408 DATABASE MANAGEMENT SYSTEMS LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	ER Diagram	CO1
2	DDL Commands	CO2
3	DML Commands	CO2
4	Grouping Functions	CO2
5	Aggregate Functions in SQL	CO2
6	Set Operations	CO2
7	Views and Assertions	CO2
8	TCL Commands	CO2
9	DCL Commands	CO2
10	Temporary Table	CO2
11	PL/SQL Control Structures	CO3

12	Procedures	CO3
13	Exception Handling	CO3
14	Trigger – I	CO3
15	Trigger – II	CO3
16	Cursor	CO3

Vidya Academy of Science & Technology

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATA SCIENCE AND MINING LAB

UCEST105 ALGORITHMIC THINKING WITH PYTHON LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Demonstrate about Basics of Python Programming	CO1
2	Demonstrate about fundamental Data types in Python Programming (int, float, complex, bool, string)	CO1
3	Demonstrate different Arithmetic Operations on numbers in Python	CO2
4	Create, concatenate, and print a string and access a sub-string from a given string	CO3
5	Familiarize time and date in various formats	CO3
6	Create, append, and remove lists in Python using NumPy	CO3
7	Program to find the largest of three numbers	CO2, CO3
8	Convert temperatures to and from Celsius and Fahrenheit	CO2, CO3
9	Construct the stars (*) pattern using a nested for loop	CO2, CO3
10	Program that prints prime numbers less than 20	CO2, CO3
11	Find the factorial of a number using Recursion	CO3

12	Recursive function to add two positive numbers	CO3
13	Recursive function to multiply two positive numbers	CO3
14	Recursive function to find the GCD of two positive numbers	CO3
15	Check whether the sides of a triangle form a right triangle, implemented using functions	CO2, CO3
16	Define a module to find Fibonacci Numbers and import the module into another program	CO3, CO4
17	Define a module and import a specific function from it into another program	CO4
18	Check whether a given number is a valid mobile number, using functions	CO4

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

NETWORK & INFORMATION SYSTEMS LAB

UCEST105 ALGORITHMIC THINKING WITH PYTHON LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
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2	Demonstrate about fundamental Data types in Python Programming (int, float, complex, bool, string)	CO1
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NETWORK & INFORMATION SYSTEMS LAB

CSL411 COMPILER LAB

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8	Implementation of Calculator using LEX and YACC.	CO2
9	Convert the BNF rules into YACC form and write code to generate an Abstract Syntax Tree (AST).	CO2
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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

NETWORK & INFORMATION SYSTEMS LAB

PCCSL 307 DATA STRUCTURES LAB

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Find the sum of two sparse polynomials using arrays.	CO1,C04
2	Find the transpose of a sparse matrix and the sum of two sparse matrices.	CO1,CO2,CO4
3	Convert an infix expression to postfix (or prefix) and evaluate it using a stack.	CO1,CO2,CO4
4	Implement Queue, Dequeue (Double Ended Queue), and Circular Queue using arrays.	CO1,CO2,CO4
5	Implement backward and forward navigation of visited web pages in a web browser using doubly linked list operations.	CO1,CO2,CO4
6	Implement addition and multiplication of polynomials using singly linked lists.	CO1,CO2,CO4
7	Create a binary tree for a given simple arithmetic expression and find its prefix/postfix equivalent.	CO1,CO2,CO4
8	Implement a dictionary of word–meaning pairs using a Binary Search Tree (BST).	CO1,CO2, CO3,CO4
9	Find the shortest distance of every cell from a landmine inside a maze.	CO1, CO2
10	Solve the Water Jug Problem (10 L, 7 L, 4 L containers) by modeling it as a graph and determine whether exactly	CO1, CO2

	2 litres can be obtained in the 7 L or 4 L container.	
11	Implement the Find and Replace feature in a text editor.	CO1,CO2,CO4
12	Given a sorted array, implement an efficient searching algorithm to search for a specific item.	CO1,CO2,CO3
13	Implement Bubble Sort, Insertion Sort, Radix Sort, Quick Sort, and Merge Sort and compare the number of steps involved.	CO1,CO2,CO3
14	Generate the order of serving customers in a post office based on priority: Differently Abled → Senior Citizen → Defence Personnel → Ordinary Customer.	CO1,CO2
15	Implement a spell checker using a hash table to store a dictionary and suggest corrections for misspelled words.	CO1,CO2,CO3,CO4
16	Simulate a basic memory allocator and garbage collector using a doubly linked list.	CO1,CO2,CO4
17	Given activity points for events and a maximum of k events that can be attended, determine the maximum activity points that can be earned.	CO1,CO2
18	Merge K sorted lists into a single sorted list using a min-heap.	CO1,CO2,CO3

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROGRAMMING LAB

PCCSL 507 NETWORKS LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Familiarize Linux networking commands (ifconfig, ping, traceroute, netstat, nmap, tcpdump, Wireshark, etc.)	CO1
2	Wireshark – HTTP packet analysis	CO3
3	Wireshark – SMTP packet analysis	CO3
4	Wireshark – DNS packet analysis	CO3
5	TCP Client-Server communication (Matrix type identification)	CO2
6	UDP Client-Server communication (Text translation)	CO2
7	Multi-user chat server using TCP	CO2
8	Concurrent Time Server using UDP	CO2
9	Concurrent File Server using TCP	CO2
10	Packet capturing application using Raw Sockets	CO4
11	Cisco Packet Tracer – Familiarizing router commands	CO1,CO4

12	Static Routing	CO4
13	RIP Version 2 Routing	CO4
14	OSPF Routing	CO4
15	Access Control List (ACL) configuration	CO4
16	Extended ACL – Block HTTP access to Hackathon website	CO4
17	IPv6 Routing using RIPng	CO3,CO4

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROGRAMMING LAB

UCEST105 ALGORITHMIC THINKING WITH PYTHON LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
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12	Recursive function to add two positive numbers	CO3
13	Recursive function to multiply two positive numbers	CO3
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15	Check whether the sides of a triangle form a right triangle, implemented using functions	CO2, CO3
16	Define a module to find Fibonacci Numbers and import the module into another program	CO3, CO4
17	Define a module and import a specific function from it into another program	CO4
18	Check whether a given number is a valid mobile number, using functions	CO4

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROGRAMMING LAB

PCCSL 307 DATA STRUCTURES LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Find the sum of two sparse polynomials using arrays.	CO1,C04
2	Find the transpose of a sparse matrix and the sum of two sparse matrices.	CO1,CO2,CO4
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5	Implement backward and forward navigation of visited web pages in a web browser using doubly linked list operations.	CO1,CO2,CO4
6	Implement addition and multiplication of polynomials using singly linked lists.	CO1,CO2,CO4
7	Create a binary tree for a given simple arithmetic expression and find its prefix/postfix equivalent.	CO1,CO2,CO4
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9	Find the shortest distance of every cell from a landmine inside a maze.	CO1, CO2

10	Solve the Water Jug Problem (10 L, 7 L, 4 L containers) by modeling it as a graph and determine whether exactly 2 litres can be obtained in the 7 L or 4 L container.	CO1, CO2
11	Implement the Find and Replace feature in a text editor.	CO1,CO2,CO4
12	Given a sorted array, implement an efficient searching algorithm to search for a specific item.	CO1,CO2,CO3
13	Implement Bubble Sort, Insertion Sort, Radix Sort, Quick Sort, and Merge Sort and compare the number of steps involved.	CO1,CO2,CO3
14	Generate the order of serving customers in a post office based on priority: Differently Abled → Senior Citizen → Defence Personnel → Ordinary Customer.	CO1,CO2
15	Implement a spell checker using a hash table to store a dictionary and suggest corrections for misspelled words.	CO1,CO2, CO3,CO4
16	Simulate a basic memory allocator and garbage collector using a doubly linked list.	CO1,CO2,CO4
17	Given activity points for events and a maximum of k events that can be attended, determine the maximum activity points that can be earned.	CO1,CO2
18	Merge K sorted lists into a single sorted list using a min-heap.	CO1,CO2,CO3

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

OBJECT ORIENTED PROGRAMMING LAB

UCEST105 ALGORITHMIC THINKING WITH PYTHON LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Demonstrate about Basics of Python Programming	CO1
2	Demonstrate about fundamental Data types in Python Programming (int, float, complex, bool, string)	CO1
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15	Check whether the sides of a triangle form a right triangle, implemented using functions	CO2, CO3
16	Define a module to find Fibonacci Numbers and import the module into another program	CO3, CO4
17	Define a module and import a specific function from it into another program	CO4
18	Check whether a given number is a valid mobile number, using functions	CO4

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ADVANCED COMPUTER LAB

UCEST105 ALGORITHMIC THINKING WITH PYTHON LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Demonstrate about Basics of Python Programming	CO1
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17	Define a module and import a specific function from it into another program	CO4
18	Check whether a given number is a valid mobile number, using functions	CO4

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ADVANCED COMPUTER LAB

PCCSL307 DATA STRUCTURES LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Find the sum of two sparse polynomials using arrays.	CO1,CO4
2	Find the transpose of a sparse matrix and the sum of two sparse matrices.	CO1,CO2,CO4
3	Convert an infix expression to postfix (or prefix) and evaluate it using a stack.	CO1,CO2,CO4
4	Implement Queue, Dequeue (Double Ended Queue), and Circular Queue using arrays.	CO1,CO2,CO4
5	Implement backward and forward navigation of visited web pages in a web browser using doubly linked list operations.	CO1,CO2,CO4
6	Implement addition and multiplication of polynomials using singly linked lists.	CO1,CO2,CO4
7	Create a binary tree for a given simple arithmetic expression and find its prefix/postfix equivalent.	CO1,CO2,CO4
8	Implement a dictionary of word–meaning pairs using a Binary Search Tree (BST).	CO1,CO2, CO3,CO4
9	Find the shortest distance of every cell from a landmine inside a maze.	CO1, CO2
10	Solve the Water Jug Problem (10 L, 7 L, 4 L containers) by modeling it as a graph and determine whether	CO1, CO2

	exactly 2 litres can be obtained in the 7 L or 4 L container.	
11	Implement the Find and Replace feature in a text editor.	CO1,CO2,CO4
12	Given a sorted array, implement an efficient searching algorithm to search for a specific item.	CO1,CO2,CO3
13	Implement Bubble Sort, Insertion Sort, Radix Sort, Quick Sort, and Merge Sort and compare the number of steps involved.	CO1,CO2,CO3
14	Generate the order of serving customers in a post office based on priority: Differently Abled → Senior Citizen → Defence Personnel → Ordinary Customer.	CO1,CO2
15	Implement a spell checker using a hash table to store a dictionary and suggest corrections for misspelled words.	CO1,CO2, CO3,CO4
16	Simulate a basic memory allocator and garbage collector using a doubly linked list.	CO1,CO2,CO4
17	Given activity points for events and a maximum of k events that can be attended, determine the maximum activity points that can be earned.	CO1,CO2
18	Merge K sorted lists into a single sorted list using a min-heap.	CO1,CO2,CO3

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ADVANCED COMPUTER LAB

PCCSL308 DIGITAL LAB

LIST OF EXPERIMENTS

SL.NO	NAME OF THE EXPERIMENT	CO MAPPING
1	Study of basic digital ICs using digital logic gates.	CO1
2	Design and implement a combinational logic circuit Half adder, full adder, half subtractor, full subtractor	CO1
3	Realization of an SOP and its corresponding POS expression using NAND gates alone and NOR gates alone	CO1
4	Design and implement a combinational logic circuit Code Converters	CO1
5	Realization of Verilog modules for basic gates and familiarization of Verilog	CO1
6	Design and implement Half adder, full adder using Verilog	CO1
7	Design and implement Code converters using Verilog	CO1
8	Design and implement Encoder, Decoder using Verilog	CO2
9	Design and implement Multiplexer, Demultiplexer using Verilog	CO2
10	Design and implement 3-bit Synchronous Up-Counter using Verilog	CO3
11	Design and implement D Flip Flop using Verilog	CO3

12	Familiarization of LogiSim – an educational tool for Digital Logic Circuit Design and Simulation	CO1
13	Design and implement combinational circuits using MSI devices using LogiSim	CO2
14	Implement D Flip Flop and JK Flip Flop using LogiSim	CO3
15	Implement Shift Registers using LogiSim	CO3
16	Implement 3-bit Asynchronous Up-Counter using LogiSim	CO3
17	Implement 3-bit Synchronous Up-Counter using LogiSim	CO3
18	Design and implement a finite-state machine - a serial bit sequence detector using Verilog	CO4

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SYSTEMS LAB

UCEST105 ALGORITHMIC THINKING WITH PYTHON LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Demonstrate about Basics of Python Programming	CO1
2	Demonstrate about fundamental Data types in Python Programming (int, float, complex, bool, string)	CO1
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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SYSTEMS LAB

PCCSL308 DIGITAL LAB

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16	Implement 3-bit Asynchronous Up-Counter using LogiSim	CO3
17	Implement 3-bit Synchronous Up-Counter using LogiSim	CO3
18	Design and implement a finite-state machine - a serial bit sequence detector using Verilog	CO4

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATABASE SYSTEMS LAB

UCEST105 ALGORITHMIC THINKING WITH PYTHON LAB

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12	Recursive function to add two positive numbers	CO3
13	Recursive function to multiply two positive numbers	CO3
14	Recursive function to find the GCD of two positive numbers	CO3
15	Check whether the sides of a triangle form a right triangle, implemented using functions	CO2, CO3
16	Define a module to find Fibonacci Numbers and import the module into another program	CO3, CO4
17	Define a module and import a specific function from it into another program	CO4
18	Check whether a given number is a valid mobile number, using functions	CO4

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATABASE SYSTEMS LAB

PCCSL307 DATA STRUCTURES LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Find the sum of two sparse polynomials using arrays.	CO1,C04
2	Find the transpose of a sparse matrix and the sum of two sparse matrices.	CO1,CO2,CO4
3	Convert an infix expression to postfix (or prefix) and evaluate it using a stack.	CO1,CO2,CO4
4	Implement Queue, Dequeue (Double Ended Queue), and Circular Queue using arrays.	CO1,CO2,CO4
5	Implement backward and forward navigation of visited web pages in a web browser using doubly linked list operations.	CO1,CO2,CO4
6	Implement addition and multiplication of polynomials using singly linked lists.	CO1,CO2,CO4
7	Create a binary tree for a given simple arithmetic expression and find its prefix/postfix equivalent.	CO1,CO2,CO4
8	Implement a dictionary of word–meaning pairs using a Binary Search Tree (BST).	CO1,CO2, CO3,CO4
9	Find the shortest distance of every cell from a landmine inside a maze.	CO1, CO2
10	Solve the Water Jug Problem (10 L, 7 L, 4 L containers)	CO1, CO2

	by modeling it as a graph and determine whether exactly 2 litres can be obtained in the 7 L or 4 L container.	
11	Implement the Find and Replace feature in a text editor.	CO1,CO2,CO4
12	Given a sorted array, implement an efficient searching algorithm to search for a specific item.	CO1,CO2,CO3
13	Implement Bubble Sort, Insertion Sort, Radix Sort, Quick Sort, and Merge Sort and compare the number of steps involved.	CO1,CO2,CO3
14	Generate the order of serving customers in a post office based on priority: Differently Abled → Senior Citizen → Defence Personnel → Ordinary Customer.	CO1,CO2
15	Implement a spell checker using a hash table to store a dictionary and suggest corrections for misspelled words.	CO1,CO2, CO3,CO4
16	Simulate a basic memory allocator and garbage collector using a doubly linked list.	CO1,CO2,CO4
17	Given activity points for events and a maximum of k events that can be attended, determine the maximum activity points that can be earned.	CO1,CO2
18	Merge K sorted lists into a single sorted list using a min-heap.	CO1,CO2,CO3

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATABASE SYSTEMS LAB

CSL411 COMPILER LAB

LIST OF EXPERIMENTS

SL.NO.	NAME OF THE EXPERIMENT	CO MAPPING
1	Design and implement a lexical analyzer using C language to recognize all valid tokens in the input program. The lexical analyzer should ignore redundant spaces, tabs, newlines and comments.	CO1
2	Implement a Lexical Analyzer for a given program using LEX Tool.	CO1
3	Write a LEX program to display the number of lines, words and characters in an input text.	CO1
4	Write a LEX program to convert the substring abc to ABC from the given input string.	CO1
5	Write a LEX program to find the total number of vowels and consonants from the given input string.	CO1
6	Generate a YACC specification to recognize a valid arithmetic expression that uses operators +, -, *, / and parenthesis.	CO2

7	Generate a YACC specification to recognize a valid identifier which starts with a letter followed by any number of letters or digits.	CO2
8	Implementation of Calculator using LEX and YACC.	CO2
9	Convert the BNF rules into YACC form and write code to generate an Abstract Syntax Tree (AST).	CO2
10	Write a program to find ϵ -closure of all states of any given NFA with ϵ transition.	CO3
11	Write a program to convert NFA with ϵ transition to NFA without ϵ transition.	CO3
12	Write a program to convert NFA to DFA.	CO3
13	Write a program to minimize any given DFA.	CO3
14	Write a program to find FIRST and FOLLOW of any given grammar.	CO4
15	Design and implement a Recursive Descent Parser for a given grammar.	CO4
16	Construct a Shift Reduce Parser for a given language.	CO5
17	Write a program to perform constant propagation.	CO5
18	Implement Intermediate Code Generation for simple expressions.	CO5
19	Implement the back end of the compiler which takes the three-address code and produces 8086 assembly language instructions (move, add, sub, jump, etc.).	CO5